

TIRUMALESH NAGOTHI

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EDUCATION

M.S. Computer Science, Illinois Institute of Technology, College of Computing, Chicago, IL. GPA: 3.5/4.0 Jan 2023 - Dec 2024
B.Tech. Computer Science and Engineering, GITAM University, Visakhapatnam, India. GPA: 3.21/4.0 Jun 2018 - Apr 2022

TECHNICAL SKILLS

AI and LLMs: AI agent development, agentic workflows, LLM integration, OpenAI SDK, Anthropic Claude API, LangChain, RAG pipelines, prompt engineering, tool and function calling, embeddings and vector search, local AI models (Ollama, llama.cpp), MCP-style integrations

Languages: Python, TypeScript, JavaScript, SQL

Frontend: React, React Native, Next.js, Tailwind CSS, REST and WebSocket clients, responsive UI

Backend and APIs: FastAPI, Flask, REST APIs, WebSockets, microservices, JWT authentication, OAuth 2.0, role-based access control

Data and ML: Pandas, NumPy, Scikit-learn, PySpark, ETL pipelines, data processing

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, caching

Cloud and DevOps: AWS (EC2, S3, Lambda, RDS, DynamoDB, EventBridge, CloudFront), Docker, NGINX, serverless architectures, CI/CD pipelines, GitHub Actions, Git, Linux, logging and monitoring

PROFESSIONAL EXPERIENCE

Full Stack AI Engineer | Prozech Solutions | Remote, USA Feb 2026 - Present

- Build AI agent workflows with LangChain, chaining retrieval and tool calls into multi-step automation pipelines that replace manual steps in internal monitoring and operations processes.
- Build the LLM integration layer connecting Claude and the OpenAI SDK to FastAPI backend services for document parsing and structured data extraction, with schema validation, retries, and fallback handling.
- Build a real-time analytics dashboard in TypeScript against REST APIs, with LLM-driven summarization served by locally hosted models to keep sensitive data in the environment.
- Persist AI agent state across runs on AWS Lambda and DynamoDB, with logging and monitoring on every step of the workflow.
- Ship on AWS (EC2, S3, Lambda) with JWT authentication, PostgreSQL and Redis, and Docker containerization behind CI/CD.

Founding Software Engineer | Rivalarc Inc. | Chicago, IL Sep 2024 - Jan 2026

- Built a cross-platform mobile product end to end and released it on the App Store and Play Store: React Native frontend, FastAPI backend, MongoDB, and AWS infrastructure.
- Designed REST and WebSocket APIs for faceoffs, real-time voting, tournaments, and leaderboards, secured with JWT, OAuth 2.0, and role-based access control.
- Built a vote integrity system to detect duplicate and coordinated voting, and a content moderation pipeline on AWS Lambda and EventBridge for user-uploaded media, with S3 storage and CloudFront delivery.
- Tuned MongoDB schemas and aggregation pipelines and added Redis caching to bring API response times down and keep real-time sync smooth across iOS and Android.
- Containerized the backend with Docker and NGINX, set up Git-based CI/CD on AWS, instrumented logging and alerting, and iterated on features end to end based on real usage.

Application Developer (Contract) | Leap of Faith Technologies | Libertyville, IL Jun 2024 - Aug 2024

- Built full-stack features for a healthcare application covering Medicare MIPS reporting and clinical data management: TypeScript frontend, Python and FastAPI backend, REST APIs.
- Wrote Python pipelines with Pandas and NumPy to validate and transform clinical data from multiple upstream sources, feeding PostgreSQL-backed reporting dashboards.
- Integrated third-party LLM APIs for text analysis and structured extraction with error handling and fallback paths; shipped via Docker in Agile sprints alongside senior engineers.

ML Engineer | TCR Innovation | Mumbai, India Jun 2021 - Nov 2022

- Built and tuned classification models (logistic regression, random forest, SVM) in Python for cancer diagnosis prediction; gains came from data cleaning and feature engineering.
- Built automated ETL pipelines with Pandas, NumPy, and PySpark, exposing results through REST endpoints backed by PostgreSQL and MySQL for internal analysts.
- Containerized models with Docker on AWS EC2 and built a Flask inference API with versioning and monitoring.

PROJECTS

- **DataTellsDS (open source).** No-code AI/ML automation platform taking a raw dataset through profiling, model selection, and training to a deployable inference endpoint. Python, Flask, React and TypeScript, JWT auth, role-based access, offline model execution, Docker, CI/CD.
- **Local Knowledge Vault.** Markdown knowledge base with a custom query interface doing retrieval over personal notes against a locally hosted LLM, with no external API calls. Python, local inference.

RESEARCH EXPERIENCE

Research Assistant | GITAM University | Visakhapatnam, India Jan 2022 - Mar 2022

- Built an ML-based institution rank prediction system using Scikit-learn classification models, with a recommendation layer surfacing the indicators an institution could act on to improve its future ranking; published in the Journal of Algebraic Statistics, Vol. 13 No. 3 (2022) [\[LINK\]](#)

CERTIFICATIONS

IBM Data Science Professional Certificate, Coursera (B7FYCHXB4UX) | Google Data Analytics Professional Certificate, Coursera (PBYB9NS3J6P2)